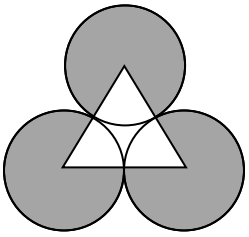
SECTION 3: MATHEMATICS PROFICIENCY
Time -- 75 minutes
60 Questions (146 – 205)

NAME: _____
SCHOOL: _____
SIMULATED EXAM – SET A

For each question, decide which the correct answer is, and darken the corresponding oval on the answer sheet.

146. Given a linear function with y-intercept of 3 and passes through $(4, -3)$, find the zero of a function?
A. 0
B. 2
C. 3
D. 4

147. Three five-peso coins were placed side-by-side in such a way that their centers formed an equilateral triangle. Find the area of the shaded region if the radius of five-peso coin was 1 cm.



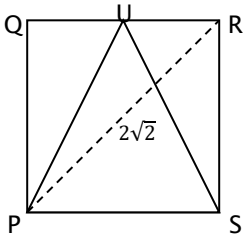
A. 2π cm
B. 2.5π cm
C. 3π cm
D. 3.5π cm

148. Find the value of a.

$$\frac{a \cdot a \cdot a}{a + a + a} = 12$$

A. 6
B. 12
C. ± 6
D. ± 12

149. ΔPUS is inscribed in $\square$PQRS such that U is the midpoint of QR. If the length of the diagonal of the square is $2\sqrt{2}$, what is the perimeter of ΔPUS?
A. $2(\sqrt{5} + 1)$ units
B. 5 units
C. $1 + 2\sqrt{5}$ units
D. 4 units

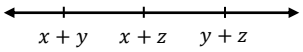


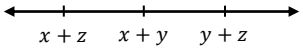
150. The ratio of the angles of a certain quadrilateral is $1:2:4:5$. What is the measure of the largest angle?
A. $110°$
B. $120°$
C. $140°$
D. $150°$

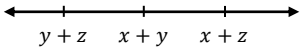
151. If k is an odd integer, which of the following is an even integer?
A. k^2
B. $k^2 + 2$
C. $k^2 + 2k$
D. $k^2 + 2k + 1$

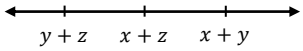
152. Which of the following is a function?
A. $|x + y| = 1$
B. $|x| + y = 1$
C. $|x| + |y| = 1$
D. $x + |y| = 1$

153. If x, y, and z are integers, where $x < y < 0 < z$, which of the following is true?

A. 

B. 

C. 

D. 

154. If x and y are both negative integers, which of the following is not possible?

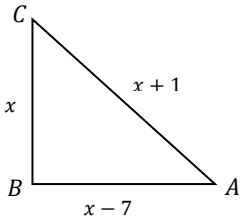
A. $x - y < 0$

B. $x + y < 0$

C. $-xy < 0$

D. $xy < 0$

155. In the figure, ΔABC is a right triangle. What is the value of x?



A. 4

B. 5

C. 12

D. 13

156. A mother baked a cake. She gave one-third of it to her husband, one-fourth to her son and two-fifths of the remaining to her daughter. What part of the cake was left?

A. $\frac{2}{3}$

B. $\frac{1}{3}$

C. $\frac{1}{60}$

D. $\frac{1}{4}$

157. If $\log_b x = y$ corresponds to $b^y = x$, then find the value of $\log_4 8$.

A. $\frac{3}{2}$

B. 4

C. 2

D. $\frac{1}{2}$

158. Find the reciprocal of $+\frac{1}{b}$.

A. $\frac{b}{ab+1}$

B. $\frac{b}{a+b}$

C. $\frac{b}{a+1}$

D. $\frac{b}{a+b}$

159. Simplify the expression $6^{x+1} \cdot 6^{x-1}$.

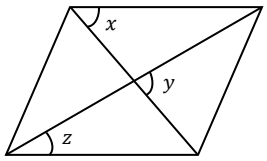
A. 6^{2x+2}

B. 6^{x^2}

C. 6^{x^2-1}

D. 6^{2x}

160. Given a parallelogram, find the measure of angle z in terms of x and y.



A. $x + y$

B. $x - y$

C. $y - x$

D. $-x - y$

161. How many times does a line pass through a parabola?

A. exactly twice

B. at least once

C. at most once

D. at most twice

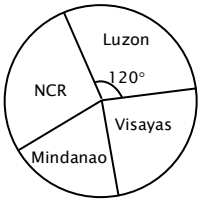
162. Given a pie chart that shows the total population of the Philippines. If the total population is 90 million, how many are from Luzon?

A. 22.5 million

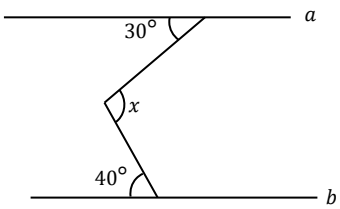
B. 25 million

C. 30 million

D. 75 million



163. In the figure $a \| b$, Find the measure of $\angle x$.



A. $35°$

B. $50°$

C. $70°$

D. $110°$

164. If r, s, and t are integers and s is a factor of t, which of the following is integer?

A. $\frac{st}{r}$

B. $\frac{sr}{t}$

C. $\frac{rt}{s}$

D. $\frac{t}{sr}$

165. What can x be in the expression $\sqrt{-x}$ so that the expression will be a real number?

A. x can be a positive integer

B. x can be a negative integer

C. x can be an imaginary number

D. x can be any real number

166. What is the value of x in $|x+1| > 0$ that will satisfy the condition?

A. all real number except -1

B. numbers that are greater than 1

C. numbers that are less than 1

D. all real numbers

167. James went to mall to buy items worth P66.00 that were not subjected to tax. However, he also bought items worth P540.00 that were still to be subjected to 12% VAT. How much did he pay all in all?

A. P606.00

B. P663.73

C. P670.80

D. P678.72

168. How much water should Ron add to 24% concentration of 50 ml orange juice to make it 10% concentration?

A. 50 ml

B. 60 ml

C. 70 ml

D. 80 ml

169. In an hour-long debate, two candidates both answered 8 questions each. Each question took 3 minutes to answer. At the end of the debate, each candidate was also given 2 minutes to give their closing statement. How long did each candidate speak?

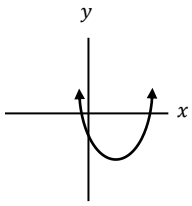
A. 30 minutes

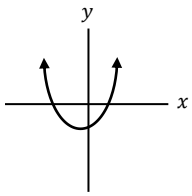
B. 32 minutes

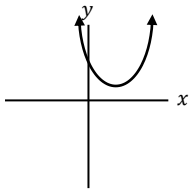
C. 26 minutes

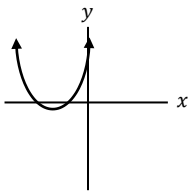
D. 60 minutes

170. Given quadratic function $y = ax^2 + bx + c$ where $a \neq 0$ and $b^2 - 4ac < 0$. Which of the following is the possible graph of the given quadratic function?

A. 

B. 

C. 

D. 

171. If $(3 \times 10^a)(0.007 \times 10^b) = (2.1 \times 10^9)$, what is the value of $a + b$?

A. 9

B. 11

C. 13

D. 15

172. If the perimeter of a circle is equal to the perimeter of a square. Which of the following is true about their area?

A. Their areas are equal.

B. The area of the square is greater than the area of the circle.

C. The area of the circle is greater than the area of the square.

D. Cannot be determined

173. If k is an odd number, which of the following will give an even answer?

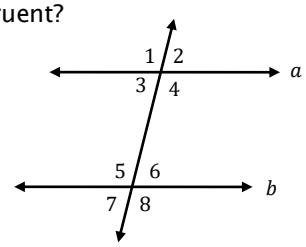
A. $k^2 + 2k$

B. $k^2 - 4k + 4$

C. $k^3 - 8$

D. $k^3 + k^2 - 2k - 2$

174. In the figure, line a is parallel to line b. Which of the following are congruent?



A. 1, 4, 6, 7

B. 1, 2, 7, 8

C. 3, 4, 5, 6

D. 2, 3, 6, 7

175. Simplify: $\frac{x^2-3x+2}{x^2-5x+6}$

A. $\frac{x-1}{x-3}$

B. $\frac{x-2}{x-5}$

C. $\frac{x+1}{x+3}$

D. $\frac{x+2}{x-5}$

176. The average of 3 numbers is N. The first number is $4N$ and the second number is $2(N-1)$. What is the third number?

A. $3N - 2$

B. $2 - 3N$

C. $1 - 3N$

D. $3N + 1$

177. One paper costs 10 cents. For every 1000 pieces of paper purchased, you get 3% discount. How much did you save, if you purchased 5000 pieces of paper?

A. P15.00

B. P150.00

C. P350.00

D. P485.00

178. If z varies jointly as x and square of y, what happens to z if x is quadrupled and y is halved?
A. half
B. double
C. triple
D. same

179. What is the slope of the line whose equation is $\sqrt{2}x + \sqrt{2}y = 7$?
A. $-\sqrt{2}$
B. $\frac{\sqrt{2}}{2}$
C. -1
D. $-\frac{\sqrt{2}}{2}$

180. If x is equal to 15 more than 3 times y and z is equal to 3 times x, then what is the relationship between y and z?
A. z is 5 more than y
B. y is 5 less than one ninth of z
C. y is 5 more than z
D. z is 45 less than 9 times of y

181. If $x^2 - y^2 = 77$ and $x - y = 11$, what is the value of x?
A. 7
B. 9
C. 11
D. 13

182. A is 60% of B. B is 60% of C. Which of the following is true?
A. A is 36% of C
B. A is 40% of C
C. A is 60% of C
D. A is 120% of C

183. Find the ratio between the radii of two spheres, if the ratio of their volume is $1:8$.
A. $3:8$
B. $1:2\sqrt{2}$
C. $1:2$
D. $1:4$

184. Given the pattern:

3	12	4	1
8	2	6	x

Find the value of x.
A. $\frac{3}{2}$
B. 4
C. 10
D. 24

185. Charlayne wants to invest part of $\$40{,}000$ at 16% simple interest rate and the remainder at 8% simple interest rate. How much should she invest at 16% to earn a total of $\$4{,}000$?
A. $\$10{,}000$
B. $\$15{,}000$
C. $\$25{,}000$
D. $\$30{,}000$

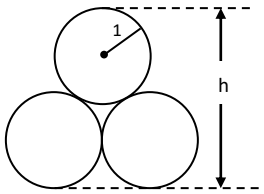
186. There are 3 boys in the class room when the window broke. All of them were asked to give a statement:
Dennis: All of us are telling the truth.
Diego: Dennis or Don is telling the truth.
Don: at least one of us is lying.

Who was/were telling the truth?
A. Dennis
B. Diego
C. Don
D. Diego and Don

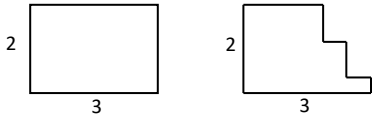
187. The average of five scores is 10. When a new score x is added, the average becomes 12. What is the value of x?
A. 10
B. 20
C. 22
D. 24

188. A bag contains different color of marbles, in which 2 are yellow, 4 are red and 4 are green. If two marbles are drawn from the bag at random and the first was yellow, what is the probability that it will be green on the second draw?
A. $\frac{2}{5}$
B. $\frac{3}{5}$
C. $\frac{1}{3}$
D. $\frac{4}{9}$

189. Given three congruent circles that are externally tangent to each other. If the radius of the circle is 1 cm, find the height (h) of the given figure.
A. 3 cm
B. $2\sqrt{3}$ cm
C. $(2+\sqrt{3})$ cm
D. 4 cm



190. Which is true regarding the two figures? Assuming consecutive sides are perpendicular.



A. The first has a greater perimeter.
B. The second has a greater perimeter.
C. They both have the same perimeter.
D. cannot be determined

191. Find the extraneous root of the radical equation:
$$\sqrt{3x+6}+\sqrt{x+3}=1$$
A. -1
B. 1
C. -2
D. 2

192. Given a circle with radius 8. Find the length of the chord 4 units away from the center.
A. $2\sqrt{3}$
B. $4\sqrt{3}$
C. $6\sqrt{3}$
D. $8\sqrt{3}$

193. Chaz has 5 times as many chocolates as Zachary. After she gave 20 chocolates to Zachary, they have now equal number of chocolates. How many chocolates do they have altogether?
A. 30
B. 40
C. 50
D. 60

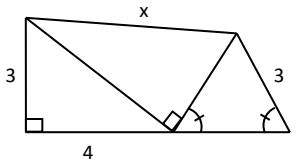
194. The length of the rectangle is 3 units more than its width. When the width is reduced by 25% and the length is increased by 3 units, the area will still be the same. Find the original dimensions.
A. 2×5
B. 4.5×7.5
C. 6×9
D. 12×15

195. The width of the rectangle is 2 less than twice of one half of the length and the perimeter is 40 cm. What is the length of the rectangle?
A. 7 cm
B. 9 cm
C. 11cm
D. 13 cm

196. What is the minimum value of the function $y = x^2 - 6x + 5$?
A. -4
B. -5
C. 8
D. 14

197. If $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 1$, find $f\big(g(2)\big)$.
A. ± 2
B. 2
C. $\pm\sqrt{2}$
D. $\sqrt{2}$

198. Refer to the figure below. Find the value of x.

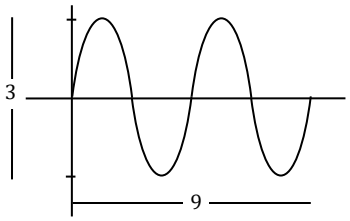


A. 4
B. $3\sqrt{3}$
C. $\sqrt{29}$
D. $\sqrt{34}$

199. Given the data 4, 6, –1, 15, –5, –2, 17, and 6. Which of the following is false?
A. The mean and the median are the same.
B. The mode is 1 more than the mean.
C. The range is 2 more than twice the median.
D. The range is 2 more than twice the sum of the mean and median.

200. *Joey's Talyer* is having a car wash service with a price of P125 per car. How many cars does it need to wash to reach their goal amount of P12,800?
A. 100
B. 101
C. 102
D. 103

201. What is the amplitude of the given graph below?



A. 1.5
B. 3
C. 9
D. 11

202. There are 20 tables and 92 chairs. Each table must only be filled with 4 or 6 chairs. How many tables are filled with 4 chairs, if all the tables and chairs are used?
A. 4
B. 6
C. 10
D. 14

203. A is odd. B is even. Which of the following must be even?
A. $A + B$
B. $4A - 3B$
C. $9AB + A$
D. $A^2 + B^2$

204. If the equation of a line is $y = mx + b$, then the slope of the line perpendicular to the given line is _____.

A. $-\frac{1}{m}$

B. $-\frac{1}{b}$

C. m

D. b

205. Chelsey was given a 20% discount on the selling price of an item. He was charged a 10% tax on the discounted price. Which of the following must be true?

A. The amount Chelsey paid was 15% lower than the original price.

B. The discount had no effect in the original price because of the tax.

C. The original price was 10% higher than the amount Chelsey paid.

D. The amount Chelsey paid was 88% of the original price.